Violet **Ross**

Computer Science at CU Boulder

Boulder, CO

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Education

University of Colorado Boulder — *Boulder, CO*

PhD in Computer Science — *Aug. 2025 - Present*

- 3.94 GPA

Middlebury College — *Middlebury, VT*

B.A. in Computer Science and Mathematics — *Sep. 2021 - May 2025*

- 3.96 GPA
- Phi Beta Kappa, Summa Cum Laude, Highest Honors Computer Science, High Honors Mathematics

Experience

Research

University of Colorado Boulder — *Boulder, CO*

Computer Science Research Assistant — *Aug. 2025 - Present*

- Conducting mathematical methods research in network science and computational infectious disease epidemiology.

Middlebury College — *Middlebury, VT*

Computer Science Research Assistant — *Jun. 2024 - Aug. 2024*

- Designed a growth model for labelled hypergraphs and implemented it in Python.
- Utilized probabilistic and analytical methods to describe the relationship between parameter choice and the distribution of node labels.
- Verified these results by carrying out hypergraph growth simulations on Middlebury's high-performance computing cluster.

Yale University — *New Haven, CT*

Mathematics Research Assistant — *Jun. 2023 - Jul. 2023*

- Participated in Yale University's "SUMRY" REU in mathematics on the Ambient Dimension of Manifold-fitting Data project.
- Implemented a model in R to non-trivially embed an intrinsically 3-dimensional manifold in any ambient dimension and then uniformly sample points from the embedding.
- Conducted research on the impact of high ambient dimension on the accuracy of intrinsic dimension estimators through both experiments and theoretical methods.

Middlebury College — *Middlebury, VT*

Statistics Research Assistant — *Jun. 2022 - Aug. 2022*

- Quantified social media responses to U.S. drone strikes in Yemen.
- Published a paper on these findings and gave talks at the 2023 Joint Mathematics Meetings and the 2023 Joint Statistical Meetings.

Teaching

Middlebury College — *Middlebury, VT*

MATH 0118: Introduction to Data Science Teaching Assistant — *6 Semesters*

- Provided daily in-class programming and debugging support for students.
- Held two additional help sessions per week.
- Graded all homework assignments.

Middlebury College — *Middlebury, VT*

CSCI 301: Theory of Computation Teaching Assistant — *1 Semester*

- Held two help sessions per week for students.

Middlebury College — *Middlebury, VT*

MATH 0218: Statistical Learning Teaching Assistant — *1 Semester*

- Held one help session per week for students.

Middlebury College — *Middlebury, VT*

MATH 105: Exponents, Logarithms, and Functions (Pre-Calculus) Teaching Assistant — *1 Semester*

- Provided daily in-class problem-solving support to students.

Publications

Investigating responses to US drone strikes in Yemen using Twitter data — *Media, War, & Conflict*

Evan Weiss, Violet Ross, Alex Lyford — *2024*

In Revision

Growing Hypergraphs with Homophily — *arXiv*

Violet Ross, Francis Cataldo, Philip S. Chodrow — *2026*

In Progress

Efficient Sampling for Modeling Viral Kinetics

Violet Ross, Daniel B. Larremore — *2026*

Talks

Colorado College Seminar Series — *Colorado Springs, CO*

"Growing Hypergraphs with Homophily" — *Nov. 2026*

- Invited lecture

International School & Conference on Network Science — *Boston, MA*

"Modeling Homophilic Hypergraph Growth Using Edge Copying" — *Jun. 2026*

- Awarded Best Student Talk

Joint Mathematics Meetings — *Boston, MA*

"Quantifying Responses to U.S. Drone Strikes in Yemen Using Twitter Data" — *Jan. 2023*

Joint Statistical Meetings — *Toronto, Canada*

"Characterizing the Communities that Employ Academic Tracking Across the U.S." — *Aug. 2023*

Middlebury College Introduction to Data Science Course — *Middlebury, VT*

Invited Guest Lecturer — *Oct. 2022, Apr. 2023, Apr. 2024, Oct. 2024, Apr. 2025*